

NDNSF: A DATA-CENTRIC SERVICE FRAMEWORK FOR SECURE, COLLABORATIVE,
AND DYNAMIC NETWORKED APPLICATIONS

by

Tianxing Ma

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Academic Advisor:

Dr. Lan Wang

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The University of Memphis

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Abstract

A requester may know the required service without knowing which authorized Providers are currently available and willing to execute it. The initial NDN SF prototype supports discovery and selection of one Provider per invocation over Named Data Networking (NDN), but not multi-role collaboration. This proposal evaluates that foundation and investigates its extension to dependent service roles.

In the evaluated design, attribute-based encryption (ABE) protects discovery descriptors without enumerating Provider identities. ABE-backed challenge verification, signature validation under trust rules, and request-state checks govern service authorization. Service permissions remain independent of identity-signing credentials. Authorized but unselected candidates can read descriptors; request-scoped keys protect selected input and result exchanges. The proposed collaboration extension uses eligible positive acknowledgment (ACK) Data to assign roles and derive local dependency checks. Providers are intended to accept intermediate Data only from assigned predecessors over declared edges, whether retrieved from producers or caches. Unmanned aerial vehicle (UAV) multi-view sensing and distributed inference provide application settings for testing the extension.

Existing evaluations support limited invocation and execution-efficiency findings, not the effectiveness of the proposed collaboration mechanism. Planned evaluation compares complete authorization alternatives, tests role and dependency enforcement during execution and recovery, and measures communication, processing, and management costs. The research aims to establish when the invocation design and its collaboration extension are useful, and when their costs outweigh their benefits.

Keywords: Named Data Networking, service authorization, provider discovery, collaboration, distributed inference

Table of Contents

Chapter	Page
1 Introduction	1
1.1 Service Discovery and Authorized Invocation	1
1.2 The Initial NDNFS Foundation	1
1.3 Extending Invocation to Multi-Provider Collaboration	2
1.4 Thesis and Research Questions	2
1.5 Contribution and Evidence Boundary	3
2 Related Work and Design Alternatives	4
2.1 Named Computation, Invocation, and Composition	4
2.2 Comparing Authorization Designs under the Same Requirements	4
2.2.1 DNMP as a Concrete Trust-Schema Design	4
2.2.2 Adapting the Alternative to NDNFS	5
2.3 Protected Discovery and Permission Management	6
3 NDNFS System Model and Protocol	7
3.1 Participants, Trust, and Exclusions	7
3.2 Four Named Message Types	8
3.3 Authorization and Confidentiality	9
3.3.1 Rationale for ABE-Backed Service Authorization	9
3.3.2 Authorization Mechanisms, Lifecycle, and Costs	12
3.4 Proposed Extension: From ACKs to a Collaboration Plan	13
3.5 Proposed Dependency Admission and Execution State	14
3.6 Conditional Correctness Argument	15
4 Application Validation	17
4.1 UAV: Multi-View Collection, Processing, and Return	17
4.2 Distributed Inference: Explicit Data and State Dependencies	18
4.3 Current Scope and Completion Gates	19
5 Available Evidence and Required Evaluation	20
5.1 Measured Findings Are Configuration-Specific	20
5.2 Evidence Needed for Each Research Question	22
6 Research Plan, Risks, and Decision Gates	23
6.1 Scope Before More Features	23
6.2 Limits That Must Remain Visible	24
7 Conclusion	25
References	25

Chapter 1

Introduction

1.1 Service Discovery and Authorized Invocation

Named Data Networking (NDN) supports retrieval of named Data and signature validation under trust rules [1, 2]. Sync announces dataset changes so applications can retrieve available Data [3, 4]. These mechanisms do not, by themselves, select a service executor, establish invocation permission, or guarantee completion before a deadline.

A requester may know the required service without knowing which authorized Providers are currently reachable, capable, and willing to execute it. Even when one Provider suffices, candidate discovery, authorization, selection, and result retrieval remain application responsibilities. NFN, NFaaS, RICE, and NSC support named computation or invocation [5–8]; gRPC supports extensible name resolution [9]. This research examines how NDNSF organizes these service-level decisions and what they cost, rather than assuming that other frameworks cannot discover services.

1.2 The Initial NDNSF Foundation

The initial NDNSF prototype selects one Provider from multiple candidates for each invocation. The User publishes Request Data, collects acknowledgment (ACK) Data, and publishes Selection Data for the chosen Provider, which executes and publishes Response Data. Candidates offer alternatives for the same invocation; the initial prototype does not assign them to different, interdependent roles.

Discovery and authorization remain research subjects independently of collaboration. A discovery descriptor supplies feasibility information, such as deadlines and resource requirements; application input supplies execution material. In the evaluated design, attribute-based encryption (ABE) protects descriptors, while protected-challenge verification, signatures under trust rules, and request-state checks govern authorization. Authorized but

unselected candidates can read descriptors. The study also evaluates input/result protection, permission withdrawal, and management costs.

1.3 Extending Invocation to Multi-Provider Collaboration

Some tasks require complementary services from different Providers. Independently invoking each service does not by itself specify which Provider fills each role or which intermediate objects a dependent service may accept. This proposal investigates an extension that makes those assignments and dependencies explicit while retaining the invocation foundation.

For example, nearby unmanned aerial vehicles (UAVs) could collect different views of a ground area for analysis by a compute Provider. The extension would assign sensing and processing roles and check each image against its declared dependency. Area and deadline information may be needed before ACK to assess feasibility; processing images can be disclosed through Selection. Descriptor encryption limits reading to authorized candidates, not exposure of names or traffic patterns. This case motivates the extension; recognition algorithms remain an application choice.

1.4 Thesis and Research Questions

The thesis to be evaluated is that named, verifiable objects can support runtime service discovery and authorized single-Provider invocation, and that explicit role and dependency rules can extend this foundation to multi-Provider tasks. The study evaluates the discovery/authorization design and the collaboration extension separately. Benefits must justify their discovery, validation, communication, and key-management costs under the tested conditions.

RQ1. How can NDNFSF discover willing Providers while keeping discovery descriptors confidential from unauthorized readers and checking service permissions before execution, and at what cost relative to complete alternative designs?

RQ2. How can the single-Provider invocation foundation be extended so Providers selected at runtime follow task roles and named dependencies, whether intermediate Data is retrieved from producers or caches?

RQ3. Under what workload and network conditions do these mechanisms improve completion, latency, or execution efficiency, and when do their costs outweigh their benefits?

RQ1 evaluates the invocation foundation; RQ2 investigates its collaboration extension through UAV and distributed-inference dependencies; RQ3 measures both scopes separately, including their applicable failure behavior.

1.5 Contribution and Evidence Boundary

The intended contributions are an evaluated discovery/authorization design, a collaboration protocol enforcing runtime role assignments through named-dependency checks, and evidence of their benefits and costs. These build on established NDN, encryption, and computation mechanisms.

Existing invocation experiments do not validate a multi-role plan. The extension adds role and dependency rules without replacing existing request-ID isolation. Later implementation work is not attributed to the initial prototype; collaboration end-to-end evaluation remains incomplete. Chapter 5 separates available findings from required evidence.

Chapter 2

Related Work and Design Alternatives

2.1 Named Computation, Invocation, and Composition

NFN studies named computation and expression evaluation; NFaaS studies execution placement in a named-function environment [5, 6]. RICE provides an invocation handshake with support for authentication and security information [7]. NSC specifies signed invocation Interests validated using an application-provided Trust Schema, and discusses protected input and result exchange [8]. These designs establish relevant invocation and authorization mechanisms. Experimental comparisons below refer to the tested NSC configuration, not every feature of its full design.

For the collaboration extension, the question is how a request gathers current willingness and feasibility information before assigning execution roles, and how that assignment constrains intermediate-object use. Long-term service permission does not identify the Provider assigned to a particular predecessor role. Neither a request ID nor a valid signature alone expresses that assignment. The proposed contribution is the enforced connection between runtime assignment and object use, rather than the manifest format itself. Service-composition work such as CaSCON [10] is a relevant comparator; the research will examine assignment timing, dependency validation, and recovery semantics alongside workflow-security and capability-based designs. Novelty beyond this proposed scope remains to be established by that comparison.

2.2 Comparing Authorization Designs under the Same Requirements

2.2.1 DNMP as a Concrete Trust-Schema Design

The Distributed Network Measurement Protocol (DNMP) uses role-constrained signing to authorize measurement commands [11]. Its Network Key anchors the signing hierarchy for role keys, device keys, and the distributed schema. Given `target`, `probeType`, `probeArgs`, and the user's `roleType`, the client finds a schema-permitted signing key among the user's

available keys. It rejects a missing match and selects the least privileged key if several match. The receiving Network Observer Daemon (NOD) dispatches a probe only after publication validation under the schema. Thus, this is command authorization, not merely verification of an arbitrary valid signature. DNMP also bounds publication lifetimes and retains recent publication state for replay protection, assuming loosely synchronized clocks.

2.2.2 *Adapting the Alternative to NDNFSF*

For comparison, we propose retaining the NDNFSF Request–ACK–Selection–Response workflow while replacing ABE-backed invocation checks with role-constrained signing rules. The alternative must authorize both User invocation and Provider participation, match messages to active requests, and permit execution only after Selection. These adaptations are not features attributed to DNMP itself. Both designs require descriptor confidentiality before selection, recipient-scoped input/result protection afterward, and checks for withdrawn authority.

Table 1: Alternatives under the same discovery and execution requirements.

Design	Execution decision	Non-public discovery	Management cost
DNMP-inspired alternative	Validate role-constrained signatures; check invocation state and selected Provider	Combine authorization with service-group encryption or ABE for discovery	Role credentials, trust rules, status or expiry checks, and encryption keys
NDNFSF ABE-backed	Check signatures under trust rules, challenge responses, and request/authorization state	Encrypt a descriptor for a service-authorized receiver set without listing current candidates	Identity credentials, ABE policies/DKEYs, status distribution, and rekeying

Role granularity is policy-defined: a role can cover multiple actions, and credentials can be reused where the rules permit. DNMP is therefore a concrete authorization precedent, not evidence that every service needs a new signing certificate.

This comparison concerns complete designs, not measured DNMP-versus-NDNFSF performance. The invocation-state and encryption choices must be implemented and evaluated under matched requirements before claiming a cost advantage.

Recipient encryption protects selected exchanges; service-group encryption or ABE can protect discovery before participants are selected. Trust Schema alone does not provide confidentiality, but it can be combined with either mechanism.

The proposed alternative requires current revocation-status checks or certificate expiry. Deleting a certificate from a repository does not invalidate cached copies. We do not attribute NDNSF's runtime revocation protocol to DNMP.

2.3 Protected Discovery and Permission Management

NDNSF uses ABE for two related purposes: protecting discovery descriptors before Provider selection and managing service permissions independently of identity-signing credentials. The User publishes encrypted descriptors for shared retrieval, while the Controller issues policy keys defining decryption access. The enhanced NAC-ABE library provides KP-ABE support [12]. NDNSF connects these permissions to invocation by checking protected challenge values, message signatures under trust rules, and current authorization/request state. Content encryption supplies confidentiality; the combined checks govern participation and execution. Section 3.3.1 explains the reuse of permission management and its additional costs.

The comparison must include group-key or broadcast-encryption alternatives where appropriate. A stable, small group can make conventional group encryption attractive; known Providers and simple permissions can favor signed invocation with recipient encryption. ABE becomes attractive when recipients are identified by service entitlement before their current identities are known, and multiple entitlements should remain independent of identity-signing credentials. Policy size, membership churn, encryption cost, and authority availability bound this argument.

Chapter 3

NDNSF System Model and Protocol

This chapter first describes the invocation and authorization design under evaluation, then introduces the proposed collaboration extension. Later confidentiality and lifecycle refinements are not attributed to every initial-prototype path. The basic invocation selects one Provider; role assignments and named dependency edges belong to the extension.

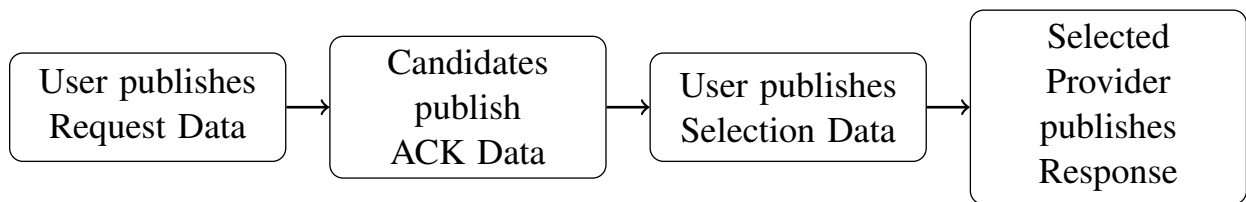
3.1 Participants, Trust, and Exclusions

These are protocol roles, not necessarily separate devices: one node may be both a User and a Provider. The managed organization registers identities and approves service permissions; the Controller administers that policy. Participants require a provisioned trust anchor and validation rules for identity certificates and Controller material. Not knowing which candidates will respond to this invocation does not mean that the authorization authority has never enrolled those entities. Billing, arbitrary business transactions, globally optimal scheduling, and protection against all denial-of-service traffic are outside the claimed scope. The User requests a service and selects its Provider; in the collaboration extension, it also coordinates role assignments. Providers offer and execute services. A Controller administers authorization material and signed status. Forwarders and caches can deliver Data without being trusted to authorize execution. The model trusts the Controller and the User's planning decision; it does not claim consensus against a malicious coordinator. An attacker may observe visible names and traffic, inject invalid objects, replay previously valid messages, or operate without the needed service permission. Credential compromise and intentional sharing by an authorized principal require additional controls and are not prevented by ABE alone.

Authorization is permission to perform an operation, authentication establishes the signing identity under trust rules, and confidentiality limits who can recover Content. Valid credentials do not prove honest load reports, correct inference, or truthful sensor observations.

Signed status must be obtained and validated before it can affect local authorization; a disconnected participant cannot be assumed to learn revocation immediately. Deadline/expiry checks require an explicit clock policy. Tests must report the allowed skew and failure behavior rather than silently assume synchronized clocks.

3.2 Four Named Message Types



Controller: authorization material and signed status.
Sync: discover names. Interests: retrieve Data.
User/Provider: validate Data, check permissions
and request state before acceptance.

Figure 1: Logical invocation dependencies. Arrows express application causality, not unsolicited network Data delivery.

Table 2: Logical message responsibilities; this is not a wire-format listing.

Data	Purpose	Principal checks
Request	Publish service requirements, minimized discovery descriptor, request identity, and Requester Challenge	Validate signature and current authorization context; recover the protected descriptor before evaluating willingness
ACK	Report positive or negative participation decision; a positive ACK carries selection information and Provider Challenge	User checks signer under trust rules, service permissions, request ID and Requester Challenge; only eligible positive ACKs enter Provider selection
Selection	Notify the chosen Provider and supply protected input or input references and request-scoped keys	Provider checks the User's signature under trust rules, intended Provider, challenge response, request state, and service permissions
Response	Publish completion/result metadata or references to protected results	Check authorized producer, request context, result identity, and decryption scope

The table describes the single-Provider invocation. Plan identity, role assignments, and dependency checks are added by the collaboration extension below, not by the initial four-message flow.

These are NDNF Data objects. Sync supports discovery of their names, after which Interests retrieve Data, possibly from a cache. A negative ACK is an application participation decision, not a network Nack. Not every observed Request must receive a reply: invalid, unauthorized, duplicate, or expired messages are subject to validation and state checks. A positive ACK expresses willingness; it does not itself start the selected task or guarantee later resource availability.

3.3 Authorization and Confidentiality

3.3.1 Rationale for ABE-Backed Service Authorization

NDNF uses ABE to protect discovery descriptors and reuses attribute-based service permissions in its invocation checks. The first choice addresses confidentiality before the

participating Providers are known; the second keeps service entitlements independent of identity-signing credentials. Neither implies that ABE is always preferable to role-based authorization with suitable encryption. Withdrawal and rekeying are lifecycle costs to evaluate. Execution authorization determines whether a User may invoke a service; Provider authorization determines whether an entity may offer that service. Content confidentiality limits who can read protected content. The Controller administers these service permissions; the User and Provider check them during invocation. Signature validation under trust rules and invocation-state checks remain necessary. Permission to offer a service is distinct from selection for one particular task.

Reason 1: confidential discovery before Provider selection. Before Provider selection, the User publishes a discovery descriptor containing the location, deadline, and resource requirements needed for a Provider's positive-ACK decision. For example, nearby UAVs need an observation area and arrival deadline to decide whether to participate, but do not yet need the full application input. The User encrypts this descriptor using ABE and the requested service's attribute, without enumerating Provider identities. Confidentiality is against parties lacking the required decryption capability: authorized but unselected Providers can still read the descriptor. Content encryption does not hide outer Data names or traffic patterns. In the request-scoped confidentiality path, the full application input is encrypted with a request-scoped key; Selection protects that key for the selected Provider. The requesting User retains access. A Trust Schema alone does not encrypt content; a role-authorized design can also use ABE or service-group encryption to provide discovery confidentiality.

Encrypted Request Data can be retrieved or cached without granting decryption access. This separation of delivery from disclosure also holds for other content-encryption schemes; it is not unique to ABE.

Reason 2: reusing service-permission management. Different entities may hold overlapping service entitlements. For example, one User may invoke observation and detection services, while another may invoke detection and mapping services. NDNSF can encode these service

sets in separately issued policy keys without changing the Users' identity-signing credentials. Service-scoped content remains separately encrypted; aggregating permissions in a key does not merge all services into one encryption scope.

In KP-ABE, ciphertext is associated with attributes and a decryption key (DKEY) encodes an access policy [12]. NDNF associates Provider access with /SERVICE/<service> and User access with /PERMISSION/<service>. Within one authority and ABE parameter generation, it combines an entity's service entitlements into an OR policy for its individually issued DKEY, while reusing its identity-signing credentials. Trust Schema specifies permitted relationships between Data names and signing key names; a validator applies those rules and checks the chain to a trust anchor. The schema by itself does not restrict who can read copied Content [13]. It can express fine-grained signing rules and be combined with encryption. NAC also protects distributed content through encryption and key distribution [14]. The rationale for ABE here is permission aggregation and reuse of the associated management mechanism, not exclusive support for cached ciphertext or credential reuse.

DNMP constrains command targets and actions through role signing keys and a Trust Schema [11]. Existing roles may cover several permissions; different entitlements may require changes to role assignments, credentials, or rules, not necessarily one new role per service.

The comparison therefore considers DNMP-inspired signing authorization with separate content encryption. ABE separates service-permission policies from identity-signing credentials; DNMP-inspired designs express permissions through role assignments and trust rules.

Neither design has a fixed credential cost. ABE requires DKEY issuance and updates, with costs that can grow with policy complexity. Role-based designs can reuse credentials; credential counts alone do not establish lower total cost.

3.3.2 Authorization Mechanisms, Lifecycle, and Costs

Connecting decryption capability to invocation authorization. In the ABE-backed invocation path, the User includes a Requester Challenge (`UserToken`) in protected Request content. A Provider that recovers it returns it in ACK Data; the User checks it against the active request. The Provider places its own Challenge (`ProviderToken`) in protected ACK content. The Provider checks the corresponding challenge response in Selection before starting the selected task.

These checks demonstrate access to protected challenge values, not the creation of new permissions or a formal proof of exclusive key possession. The User and Provider also validate message signatures and certificate chains under trust rules, check current service permissions, and match the request and its state. One-time state prevents a repeated message from authorizing another execution only while that state is preserved or safely invalidated on restart. Authorized key or challenge sharing is outside the claimed threat protection. This reuses attribute-based permission management rather than a separate set of role-based invocation rules. It is not cost-free: confidential discovery already needs Provider DKEYs, whereas User permission DKEYs and challenge processing are additional authorization costs. ABE confidentiality combined with DNMP-inspired role authorization remains a valid alternative.

Runtime permission withdrawal and evaluation. The current runtime implements permission withdrawal through Controller-signed authorization status and checks at invocation processing points. A participant must retrieve and validate updated status before enforcing the newly withdrawn authority. ABE parameter rotation and refreshed DKEYs protect subsequent exchanges encrypted under the new generation. The present rotation is Controller-wide, so other authorized members may also need refreshed DKEYs; it is not a zero-fan-out revocation scheme.

DNMP specifies validity bounds and replay-state handling [11]; these do not by themselves provide immediate withdrawal of an unexpired role key. For the comparison, a DNMP-inspired design can revoke the relevant credential and require execution nodes to check

signed revocation status or reject expired certificates. This invalidates authority at the validator; it does not physically retrieve distributed certificates. Both approaches require update propagation, and neither retracts previously disclosed keys or plaintext.

Focused integration tests exercise NDNSF’s runtime revocation checks, including identity-, certificate-, and service-scoped withdrawal. These tests establish behavior at the exercised runtime boundaries, not network-wide instantaneous revocation or lower cost than certificate-based alternatives. The remaining evaluation should measure status-propagation delay, rejection delay, rekeying traffic, and effects on unaffected members. Runtime status validation and key rotation therefore extend NDNSF to permission withdrawal, with explicit propagation and rekeying costs.

Discovery confidentiality applies to descriptor Content against parties lacking the required ABE decryption capability. It does not conceal the descriptor from other authorized candidates or provide traffic-analysis resistance. In the request-scoped protection design, the User encrypts application input with a symmetric key and delivers the necessary key material through Selection, encrypted for the chosen Provider’s authenticated public key. The response key is shared by the requesting User and selected Provider. Other Users cannot decrypt the protected result under the key-secrecy assumptions; the Provider necessarily knows the plaintext it produces. These are the protection requirements for the evaluated design, rather than claims about every historical implementation path.

3.4 Proposed Extension: From ACKs to a Collaboration Plan

The extension changes the selection problem from choosing one executor to assigning complementary task roles. The application supplies a role/dependency template, required services, feasibility constraints, and deadline. The template may be written manually; automatic generation is not necessary to establish the framework contribution. A replaceable planning policy selects among eligible positive ACKs after the collection window closes. Closing the window freezes the candidates observed for that decision, not all possible Providers in the

network. ACK claims can become stale, so a selected Provider still performs execution-time admission and may reject work it can no longer execute.

In the proposed design, the User signs a Plan Manifest containing the request identity, accepted candidate snapshot, exact role-to-Provider assignments, permitted named dependency edges, authorization/key scope, and execution identity. Its digest is computed over a canonical plan body that excludes the digest field itself. Providers validate that signed plan before deriving local dependency checks; a long-term certificate does not specify this task's assignments. The User must record the committed plan and reject a different plan for the same execution identity. This is a local state-machine rule under the trusted-User model, not a consequence of signing alone or protection against a Byzantine User signing two plans.

3.5 Proposed Dependency Admission and Execution State

Under the proposed acceptance rules, a downstream Provider checks that the execution is active and within its deadline before accepting a dependency. The Data signature must validate under the applicable trust rules; the request, plan, and execution identity must match the assignment; the signer must be the assigned predecessor; and the name, data format, and key scope must match a declared edge. Required segments must validate, and incomplete or malformed objects cannot satisfy an edge. The same checks apply to producer-, cache-, and repository-fetched Data. The plan references objects in their actual producers' authorized namespaces rather than republishing them under a different producer identity.

These are proposed acceptance conditions, not a claim that every production path is qualified. Network re-retrieval is not identical to repeating computation. If a task is reassigned, cancellation, fencing, and result-reuse rules must be explicit. An already computed image may remain useful when the new plan explicitly authorizes its reuse; an old object is not automatically invalid solely because it is cached. Application-defined compensation can limit side effects but does not provide distributed atomic transactions or exactly-once physical action.

The authorization design and comparative evaluation are intended to answer RQ1;

Table 3: Proposed task-state contract and required negative checks.

State/event	Permitted transition	Required rejection or bounded behavior
Collecting ACKs	Close window and validate candidate set	Late ACK does not silently alter a committed plan
Selected	Validate Selection and perform local admission	Positive ACK alone, wrong Provider, or spent challenge cannot start work
Waiting for dependencies	Execute when required authorized edges are satisfied	Wrong role, undeclared edge, incomplete object, or wrong active execution is rejected
Completed/cancelled	Return recorded status under duplicate-message policy	Duplicate publication does not implicitly repeat a side effect
Failure/reassignment	Explicitly terminate or authorize a replacement execution	Old execution must not acquire authority from the new assignment; allowed result reuse is explicit

the proposed plan/state rules and their tests are intended to answer RQ2. Neither question is answered by a design description alone. Existing invocation evidence does not establish collaboration correctness; wrong-role, wrong-edge, replay, cancellation, and direct/cache tests remain separate evaluation obligations.

3.6 Conditional Correctness Argument

Assume uncompromised signing keys, correctly enforced trust/authorization checks, and Providers that obey the specified state machine. Selection-before-execution follows only if every handler-start path checks an active assignment and unspent authorization state. Dependency integrity follows only if every input path, including a cache hit, checks the assigned producer and declared edge before satisfying the dependency. Replacement isolation additionally requires old workers and late callbacks to lose authority under the new execution identity. Restart recovery must preserve or safely invalidate the relevant one-time state; signature validity and cache FreshnessPeriod alone are insufficient.

These are conditional invariants and implementation proof obligations, not a cryptographic proof of a complete system. Tests should inject a wrong-role object within the same request, an undeclared edge from a correctly signed producer, and a late callback after replacement. Each isolates a property that a request ID alone cannot express. Application result correctness and physical side effects remain outside this argument.

Chapter 4

Application Validation

4.1 UAV: Multi-View Collection, Processing, and Return

The primary UAV case is a ground-location request, not a complete autonomous mission system. An operator chooses a target location or area on a map. The Request reveals only the service-level-protected information needed to assess coverage, viewing feasibility, sensor capability, and timing. Eligible UAVs return positive ACKs; a planner assigns complementary viewing roles and a processing role. Selection delivers the execution details and appropriate keys to the chosen Providers.

In the proposed UAV workflow, each sensing UAV signs ViewImage Data containing encrypted image content and publishes it in its own authorized namespace, with observation time and available pose/calibration metadata. Content encryption does not conceal the outer Data name. GPS helps relate platforms and target areas, but does not by itself establish line of sight, camera direction, accurate depth, or correspondence between views. The application must define capture tolerances, view adequacy, and failure handling. Different views cannot be treated as identical cropped inputs without matching or geometric reasoning.

The compute Provider retrieves the authorized images by their named dependencies, performs detection or other specified analysis, and publishes signed annotation Data. Its result references the original image Data and includes the permitted image keys in Content encrypted for the requesting User. This reuses the original signed images; it does not republish them under a false producer identity. The User retrieves those images and annotations and renders the overlay. Improved recognition or localization accuracy is not assumed: such a claim requires ground truth, a single-view control, and a specified fusion method.

MiniNDN [15] tests named-data execution under controlled connectivity; PX4 SITL [16] can test the application/autopilot interaction. Neither is a real-flight safety certification. The framework-level evaluation objective is authorized multi-view collection, processing, and return

before a deadline. Other flight-control, video, recording, and patrol features remain supporting engineering, not independent dissertation completion conditions.

4.2 Distributed Inference: Explicit Data and State Dependencies

The second workload partitions a specified model across Providers and exchanges named intermediate objects. Pipeline partitioning assigns sequential layers or layer groups; tensor partitioning divides operations/tensors and requires the appropriate communication and collectives. The terms describe different execution structures, not interchangeable drawing orientations. ONNX [17] represents computation graphs, but export alone does not implement cross-Provider collectives or prove distributed numerical correctness. Model partitioning has substantial prior work [18, 19]; the proposed NDNSF increment is runtime assignment and checked named dependency execution, not partitioning itself.

In the proposed execution, the planner combines a supported model/partition description with ACK capabilities. Providers obtain authorized artifact references, prepare their assigned fragments, and exchange activations or collective inputs over the declared edges. A repository provides managed artifact storage; the forwarding Content Store is opportunistic caching, not a persistence guarantee. Reuse must distinguish an identical model artifact from request-specific activation or execution state.

With KV caching, the model computes key/value (K/V) tensors, and the runtime retains the relevant state for later decoding steps. Reuse can employ shared buffers rather than copying the whole cache on every step [20]. Resident model weights do not replace this request-specific state. Autoregressive generation also requires the generated token as input to the next step; this is a temporal dependency, distinct from forward computation through the layers. If LLM streaming is retained in the final claim, evaluation must separate prefill, time to first token, per-token decoding, and KV storage/transfer. The claim will be limited to the model and partition configurations actually validated.

4.3 Current Scope and Completion Gates

The native implementation has focused test evidence, but complete end-to-end validation across the claimed process, device, and recovery configurations remains unfinished as of this September 2026 revision. A test confined to one process cannot establish cross-process behavior. The historical small-model campaign in Chapter 5 does not validate the newer artifact-reuse, tensor/hybrid, multi-accelerator, or large-language-model recovery paths.

The final campaign will freeze model version, precision, supported partition, device memory, network conditions, warm/cold state, and an output-comparison tolerance. It will include a single-Provider baseline when feasible. If the model cannot fit one device, distributed feasibility is meaningful, but no speedup ratio against a non-running baseline is claimed. UAV and DI jointly test the role/dependency abstraction; their task algorithms and device-specific adapters remain application responsibilities.

Chapter 5

Available Evidence and Required Evaluation

5.1 Measured Findings Are Configuration-Specific

Invocation campaigns below measure discovery, selection, and execution under their recorded configurations; they do not test a multi-role collaboration plan. The earlier DI diagnostic is separate application evidence, not proof that the initial framework supported role-aware collaboration.

The following numbers are retained from recorded campaigns, not new experiments in this revision [21]. The independent unit for comparative uncertainty is the seed/run, not each correlated request. Success-conditioned latency must be read with completion and timeout rates. A baseline label identifies the evaluated harness configuration, not every feature of the framework named in that label.

Table 4: Selected evidence and limits; values are unchanged from their records.

Campaign	Observation	Supported interpretation / limit
Ten-seed switching holdout	1,312 matched switching requests; NDNSF minus gRPC seed-p95 difference: -597 ms, 95% CI $[-992, -200]$; versus NSC: -2618 ms, CI $[-2916, -2315]$	Lower conditional tail latency in the tested four-Provider configuration, not universal mobility or success-rate superiority
Six-seed work-efficiency repeat	NDNSF: 1,798/1,800 completed, 1,800 execution starts; gRPC-PAR-4: 1,800/1,800, 7,180 starts	Fewer starts than this parallel policy; starts are not messages, bytes, CPU time, or energy
Earlier small-model DI diagnostic	Ten successful runs/layout: mean 259.4 ms distributed versus 186.8 ms single Provider; p95 283.9 versus 206.3 ms	Splitting was slower for this model/configuration; small success-conditioned sample does not establish a general tail-latency result

In the work-efficiency comparison, NSC-4 also completed 1,800/1,800 requests, with 1.002 execution starts per request (rounded). This does not establish a work-efficiency advantage over NSC. The mobility comparison uses common traces, publication times, deadlines,

and four Providers per system. The switching holdout uses seeds 72–81 and a registered reachability-based classification. Of its 1,312 aligned requests per system, NDNF completed 1,311, gRPC 1,297, and NSC 1,296. Each seed-p95 is calculated from that system’s successful requests, so the latency comparison is affected by different failure sets. Overall coverage controls using seeds 62–71 are a separate campaign, not the parent sample of this holdout. The small-model diagnostic is retained from earlier records; its raw-run/model provenance remains to be rechecked. Communication/setup overhead is a possible explanation, not an established cause.

The transition experiment reports 357/357 completions for NDNF without pre-registering Provider addresses, 0/357 for static-list baselines, and 357/357 for pre-registered baseline variants. It isolates the discovery configuration. A gRPC resolver or other discovery integration can update a candidate list [9]; the result does not prove that gRPC or NSC cannot discover a new Provider.

Historical v0.1 loss results include 599/600 completions at 1% per-link loss, 598/600 at 3%, and a best recorded 539/600 at 10%. The last is a selected historical run, not a current aggregate estimate or a matched comparison against later baseline data. The expression $1 - (1 - \ell)^4$ assumes four independent packet traversals with equal loss ℓ ; it is not the full service-failure probability after retries. Timeouts under heavy loss do not, without further interventions, identify one exclusive cause or rule out protocol/policy effects.

5.2 Evidence Needed for Each Research Question

Table 5: Claim-driven evaluation plan, not a completed-results checklist.

RQ	Comparison	Correctness / failure observation	Costs and outcomes
RQ1	ABE-backed authorization; role rules plus content encryption; suitable group-key and authorization combination	Unauthorized call never starts handler; replay, wrong challenge, revoked/expired state, status delay	Completion, discovery delay, crypto work, credential/policy size, renewal and withdrawal traffic
RQ2	Same task graph with static/dynamic assignment; controlled removal of each admission check	Wrong role, edge, name, key or execution; direct/cache parity; incomplete objects; cancellation or reassignment	End-to-end completion, validation cost, intermediate bytes, redundant work, bounded recovery
RQ3	Matched Provider counts, tasks, capacities, deadlines and failure schedules; feasible single-device control	Include failed and timed-out requests; distinguish network, admission, execution and result-retrieval failures	Completion and latency together; cold/warm costs; per-run uncertainty; conditions with no benefit

Each campaign must preserve its source/dependency/model identity, configurations, seeds, measurement start/end points, raw-run references, and analysis scripts. Scaling dimensions should follow the actual claim: candidates and services for discovery/authorization, edges and object sizes for collaboration. Ablation diagnoses a mechanism; an intentionally insecure variant is not a recommended deployment. A startup or collector failure must be diagnosed before attributing it to the harness or protocol. Preserve failed attempts; do not count an experiment that never reached its measurement boundary as a measured protocol failure rate.

Chapter 6

Research Plan, Risks, and Decision Gates

6.1 Scope Before More Features

The immediate priority is a frozen, runnable baseline for the final claims, not completion of every development feature. Select one UAV collection–processing graph and one supported DI partition first. Keep pipeline/tensor/hybrid generality, LLM streaming, and sophisticated recovery only where the implementation and evidence can support them. A negative performance result is an acceptable research finding when its conditions and costs are explained.

Table 6: Revised internal targets; defense is conditional on evidence and committee approval.

Period	Deliverable / decision gate
September 2026	Agree on three RQs and bounded claims; freeze supported application graphs and implementation/evidence baseline; resolve correctness blockers
October 2026	Execute registered correctness and matched comparison cases; reduce optional scope if an end-to-end gate remains open
November 2026	Freeze accepted results and provenance; write RQ answers, costs, and limitations; keep missing results visibly unverified
December 2026	Deliver an integrated complete draft to the advisor; no new optional feature branch in the research argument
January 2027	Advisor/committee review, revisions, defense readiness decision, and confirmation of current university submission/notice requirements
February 2027	Target defense only after required reviews and evidence closure; allow time for substantive corrections
Spring 2027	Complete approved corrections and final submission by the applicable Graduate School deadlines; confirm exact dates separately

The February 2027 defense target depends on evidence and committee approval. CS guidance calls for an advisor-approved complete dissertation to reach committee members two weeks before the defense [22]. The Graduate School lists February 5, 2027 for the graduation application and doctoral candidacy form, and March 26, 2027 for the defended final ProQuest submission [23]; these dates must be rechecked before filing. If core end-to-end evidence is missing at the January review, scope or timing will be reconsidered with the committee.

6.2 Limits That Must Remain Visible

ABE adds authority, computation, and rekeying costs; disconnected revocation has propagation limits. Candidate ACKs do not prove future capacity or honest reporting. Cache availability does not guarantee task completion. Tests confined to one process do not establish distributed failure behavior. UAV safety and perception accuracy need evidence beyond service-message delivery. Findings from selected hardware and models must be interpreted within those configurations.

Chapter 7

Conclusion

The research builds on NDNF's initial ability to discover multiple candidates and select one Provider for an invocation. It evaluates that discovery/authorization foundation and investigates an extension for complementary Provider roles and named dependencies. The intended contribution is an evaluated service protocol and design rationale, not a new encryption primitive or the first use of named computation. RQ1 asks when protected discovery and reuse of ABE capability are preferable to complete alternatives. RQ2 asks how runtime assignments become local rules for accepting named dependencies. RQ3 examines benefits and costs for invocation and collaboration separately. Existing campaigns establish limited invocation findings; the collaboration and cross-application claims still require their declared end-to-end evidence. This separation provides a concrete route from the present proposal to a defensible dissertation.

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